Supplementary Material
Beautified Is Not Forged: A Two-Sided Measurement of Face-Forgery Detectors under Beautification and Skin Smoothing

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Section, table and figure numbers without the prefix S refer to the paper.

S1 Pre-registration and what was read when

Every bar quoted in the paper was written into a dated file before the corresponding model was trained or the corresponding test source was read. Celeb-DF-B and the DFD filter set were read once per model, after the checkpoint had been chosen on validation data. The Celeb-DF-v2 filter set was used to diagnose which filters cause the damage (Sec. V-B), so it is a development source for every later design choice; the DFD filter set was built afterwards from a second filter library, and no per-filter result of it was inspected. Checkpoints are chosen on validation data only: FF++ validation macro-F1 (three-way, FF++ only), FF++ validation AUC (binary), or the mean of FF++ validation macro-F1 and balanced accuracy on held-out originals and renders of the two training services (three-way, full data). A selection rule used before this paper (F1 of the beautified class on an imbalanced validation set) picked the epoch with the lowest specificity on untouched originals; it was replaced before the models of this paper were trained, and the model it selected appears only in Tables S22 and S23.

What pre-registration does not control is the choice of what to test. The commercial service of Sec. V-A was read for more than 60 of our checkpoints and all 17 published detectors; its strongest smoothing level was singled out after reading it; the first exposure analysis on it was post hoc; and the strength range of the scripted smoothing was set while the service's edit magnitudes were on disk. A calibration script also fitted two scaling constants of the scripted eye and jaw edits on 120 untouched originals of the test service; it changes no image and the binary arms do not use it. The arms of the smoothing chain (Sec. S8) were declared one at a time, each after the previous result had been read. The DF40 list, detectors and statistics were fixed before any detector was run on it; the scoring of the exposure arms and of CADDM on DF40 was added afterwards. The fusion pairs, rules and reading were fixed before any fused score was computed. In Celeb-DF-B, the trimming of the tinted bars that the app adds when it re-frames a video to portrait was introduced after the first extraction lost 57% of the treated faces; both extractions are kept, and the paper reports the trimmed one.

S2 Cases near the class boundary

Table S1: Cases near the class boundary. "What changes" is the test the definition applies; it has not been measured (for example by face-recognition similarity or action-unit displacement) for any edit family. "Scored": whether a test set of the paper contains the case.

Case	Label	What changes	Scored
Camera processing, JPEG, video coding	neither	nothing about the person	all sets
Make-up worn at capture	neither	nothing after capture	incidental
Colour preset or filter	beautified	colour and tone	Celeb-DF-B, two filter sets
Skin smoothing, whitening (app or service, neural or not)	beautified	skin texture and tone	Celeb-DF-B, service
Face slimming, eye enlargement	beautified	smooth shape of own features	service, scripted
Generative "AI skin" that re-renders skin texture	beautified	skin texture, if shape and identity are kept	not scored
Relighting, virtual make-up transfer	beautified	appearance	not scored
Generative face restoration or super-resolution	boundary	texture; may alter identity cues	not scored
Face swap	forged	identity	FF++, Celeb-DF, DFD, DF40
Reenactment	forged	expression, motion	FF++, DF40
Lip-sync or dubbing	forged	speech, mouth motion	not scored
A part from another person	forged	identity of a part	part-level set
A part of the same person regenerated by inpainting	forged (boundary)	content of a part not captured	part-level set
Age, gender or attribute editing	forged	identity or attributes not captured	not scored (DF40 face editing)
Entire-face synthesis	forged	whole face not captured	not scored (DF40 synthesis)
Morphing of two identities	forged	identity	not scored
Beautified forgery; forgery of a beautified face	forged	identity or action	Celeb-DF-B, smoothed forgeries

DF40's entire-face-synthesis and face-editing sets exist but were not scored for this paper. The class definition reads only the forgery score $P(\text{fake})$; the beautified class of our three-way models is evaluated in Sec. S12.

S3 The published detectors

Table S2: Weights, training data and input of each published detector. On the video sources the twelve detectors trained on DeepfakeBench crops (the eight CNNs and the four CLIP detectors) are also scored on their own crop (dlib, five-point alignment, scale 1.3); SBI, UnivFD and NPR run on our crop; CADDM and LAA-Net re-align our crop with their own landmark or face models. On the retouching service and on DF40 every detector receives the same crop. UnivFD and NPR are general generated-image detectors, so the measurement covers fifteen face-forgery detectors and two general ones.

Detector	weights	training data	input
Xception, EfficientNet-B4, F3Net, SPSL, UCF, RECCE, CORE, SRM	DeepfakeBench release [1]	FF++ c23	256 px
SBI	authors' release (FFc23)	FF++ c23 reals, self-blended	380 px
Effort	authors' release	FF++ c23	224 px
Forensics Adapter	authors' release	FF++ c23	256 px, resized to 224 for CLIP
GenD (CLIP ViT-L/14)	authors' release	FF++ c23	224 px
DFD-HR	authors' release	FF++ c23	448 px
CADDM (EfficientNet-B4)	authors' release	FF++ raw	own 5-point alignment, 224 px
LAA-Net	authors' release (SBI-trained)	FF++ raw, self-blended	own crop rebuilt in our box, 384 px
UnivFD	authors' release (linear probe on CLIP)	ProGAN images (not faces)	224 px centre crop
NPR	authors' release	ProGAN images (not faces)	256 px

Table S3: Reported and reproduced. Published numbers are from DeepfakeBench (frame level, eight CNNs) and from the papers of the CLIP detectors. "DFB crop": our run of the released weights on DeepfakeBench crops (dlib frontal detector, 81 landmarks, five-point similarity alignment, scale 1.3, 256 px, 32 frames per video by linear spacing), built with the benchmark's own alignment function. "Our crop": landmark box with a 35% margin. Videos common to both frame sets (516 for Celeb-DF-v2, 332 for DFD). Our DFD set is a 332-video subset, so no published DFD number is comparable.

	Celeb-DF-v2 frame AUC		Celeb-DF-v2 video AUC		DFD frame AUC	
Detector	published	ours, DFB crop / our crop	published	ours, DFB crop / our crop	published	ours, DFB crop / our crop
Xception [2]	0.737	0.744 / 0.718	–	0.821 / 0.772	–	0.845 / 0.854
EfficientNet-B4 [3]	0.749	0.754 / 0.750	–	0.813 / 0.807	–	0.818 / 0.810
F3Net [4]	0.735	0.732 / 0.708	–	0.793 / 0.750	–	0.781 / 0.781
SPSL [5]	0.765	0.732 / 0.703	–	0.803 / 0.752	–	0.812 / 0.829
UCF [6]	0.753	0.775 / 0.744	–	0.841 / 0.788	–	0.818 / 0.822
RECCE [7]	0.732	0.744 / 0.688	–	0.827 / 0.731	–	0.839 / 0.833
CORE [8]	0.743	0.744 / 0.715	–	0.814 / 0.764	–	0.830 / 0.827
SRM [9]	0.755	0.763 / 0.717	–	0.844 / 0.778	–	0.837 / 0.830
Effort [10]	0.901	0.879 / 0.881	0.956	0.927 / 0.930	0.923	0.933 / 0.939
Forensics Adapter [11]	0.900	0.899 / 0.887	0.957	0.956 / 0.945	0.933	0.940 / 0.940
GenD [12]	–	0.900 / 0.891	0.960	0.960 / 0.954	–	0.932 / 0.934
DFD-HR [13]	0.910	0.886 / 0.879	0.960	0.951 / 0.943	0.953	0.942 / 0.943

Forensics Adapter reproduces its published Celeb-DF-v2 numbers on its own crop. Effort does not: our run gives 0.927 video AUC against 0.956 published, in line with an independent re-run of the released weights (0.932); comparisons of ours against Effort are therefore against the weaker reproduction. In the released training code of both, the checkpoint is saved when the metric on the configured test sets improves (Effort: `DeepfakeBench/training/trainer/trainer.py`, lines 377–390, called from `test_epoch`; Forensics Adapter: `trainer/trainer.py`, `save_best`, line 347); the papers do not state a selection rule. CADDM's released EfficientNet-B4 checkpoint gives a Celeb-DF-v2 video AUC of 0.937 on the authors' crop against 0.939 published; LAA-Net's released SBI checkpoint gives 0.943 against 0.954 on its own crop, which we rebuild inside our face box with a different face detector from the authors'. At video level Ours is at 0.947 on Celeb-DF-v2 and 0.978 on DFD (second seed 0.946 and 0.981). The binary control (frame AUC 0.914 and 0.961 on the filter sources' untreated frames) is above Effort with intervals excluding zero and on par with Forensics Adapter; a model trained on FF++ alone with our part-level forgeries does not reach it, so the margin of the full-data models comes from the external real data (FFHQ stills and commercial renders), not from the synthesis.

S4 Both errors in full

Table S4: All three beautification sources with intervals (frame level). Celeb-DF-B as in Table III; Celeb-DF-v2 and DFD: eleven and sixteen held-out filters on the full frame, ΔMISS and ΔFA with 95% video-cluster intervals.

	Celeb-DF-B (app presets)				Celeb-DF-v2 + 11 held-out filters				DFD + 16 fresh filters			
Detector	AUC	AUC_b	$\Delta MISS_{beauty}$	FA_b	AUC	AUC_f	ΔMISS	ΔFA	AUC	AUC_f	ΔMISS	ΔFA
Xception [2]	0.811	0.737	+10.9 [7.4, 14.1]	5.7	0.718	0.693	+1.1 [-1.0, 3.1]	+1.5 [-0.2, 3.2]	0.854	0.834	+2.0 [-0.6, 5.4]	+2.0 [0.2, 4.3]
EfficientNet-B4 [3]	0.811	0.762	+3.4 [0.9, 5.9]	7.8	0.750	0.725	+2.5 [0.2, 4.8]	+0.8 [-0.3, 2.6]	0.810	0.781	+6.3 [3.6, 9.2]	-0.5 [-1.9, 1.7]
F3Net [4]	0.798	0.731	+10.9 [7.4, 14.8]	5.4	0.708	0.688	-0.6 [-2.7, 1.6]	+1.3 [-0.8, 3.8]	0.781	0.774	-8.5 [-11.5, -5.5]	+3.6 [2.1, 5.7]
SPSL [5]	0.781	0.727	+12.4 [9.3, 15.5]	3.4	0.703	0.683	+3.1 [1.5, 4.9]	-0.2 [-1.5, 1.4]	0.829	0.837	+4.7 [3.2, 6.7]	-1.8 [-2.4, -0.9]
UCF [6]	0.841	0.777	+11.3 [7.6, 15.4]	5.7	0.744	0.697	+3.1 [1.0, 5.5]	+0.9 [-0.6, 2.3]	0.822	0.804	+5.0 [1.9, 8.0]	+0.3 [-1.6, 2.5]
RECCE [7]	0.785	0.723	+13.8 [10.7, 17.1]	4.1	0.688	0.661	+0.1 [-1.9, 2.0]	+0.9 [-0.8, 2.4]	0.833	0.808	+7.3 [4.6, 10.7]	-0.2 [-1.7, 1.3]
CORE [8]	0.821	0.752	+13.5 [10.2, 16.6]	4.0	0.715	0.693	-0.0 [-1.5, 1.9]	+0.4 [-0.8, 2.4]	0.827	0.821	-5.4 [-8.8, -2.9]	+1.6 [-0.3, 3.7]
SRM [9]	0.789	0.733	+15.3 [12.1, 18.7]	2.8	0.717	0.712	-0.1 [-1.4, 1.6]	+0.7 [-0.5, 2.1]	0.830	0.822	+5.8 [2.4, 9.7]	+0.2 [-1.2, 1.9]
SBI [14]	0.875	0.752	+21.8 [17.3, 26.1]	1.8	0.817	0.776	+8.2 [5.5, 10.4]	-0.6 [-1.5, 0.5]	0.829	0.810	+4.5 [2.7, 6.3]	+0.4 [-0.7, 1.4]
UnivFD‡ [15]	0.653	0.624	-7.2 [-10.2, -4.9]	7.6	0.662	0.622	-7.2 [-10.6, -4.2]	+7.8 [4.1, 11.7]	0.840	0.801	-10.2 [-13.5, -6.8]	+11.6 [8.0, 15.7]
NPR‡ [16]	0.503	0.498	-0.7 [-2.2, 0.4]	3.4	0.522	0.497	-38.0 [-43.2, -32.2]	+36.0 [28.7, 43.6]	0.473	0.516	-13.7 [-20.2, -8.4]	+9.4 [2.4, 15.3]
Effort [10]	0.909	0.807	+11.0 [7.7, 14.5]	5.4	0.881	0.813	+23.3 [19.9, 26.3]	-2.8 [-3.7, -1.7]	0.939	0.905	+15.1 [11.1, 18.6]	-2.3 [-3.1, -1.4]
Forensics Adapter [11]	0.909	0.828	+15.2 [11.9, 18.6]	7.9	0.887	0.814	+11.9 [8.6, 14.7]	+2.5 [0.3, 5.7]	0.940	0.924	+8.0 [5.6, 11.2]	-0.6 [-1.6, 0.4]
GenD [12]	0.924	0.807	+12.0 [8.2, 15.6]	9.9	0.891	0.856	-10.1 [-13.1, -7.1]	+9.9 [7.4, 13.1]	0.934	0.925	+3.1 [0.7, 5.8]	+1.2 [0.1, 2.2]
DFD-HR [13]	0.938	0.838	+3.6 [0.6, 6.8]	14.6	0.879	0.824	+8.0 [5.3, 10.5]	+1.6 [0.4, 3.8]	0.943	0.928	+9.1 [6.8, 11.8]	-2.0 [-2.9, -1.2]
CADDM [17]	0.886	0.762	-2.9 [-5.8, 0.1]	13.8	0.839	0.796	+3.2 [0.7, 5.7]	+3.0 [1.4, 4.7]	0.845	0.818	+3.8 [2.2, 5.9]	+0.8 [-0.4, 1.9]
LAA-Net [18]	0.889	0.698	+14.7 [10.9, 18.6]	6.0	0.851	0.799	+10.2 [7.4, 12.8]	-0.1 [-1.4, 1.1]	0.810	0.783	+6.2 [4.2, 7.7]	+0.0 [-0.9, 1.0]
Ours, FF++ only	0.938	0.872	+7.5 [4.8, 10.5]	7.8	0.898	0.851	-3.5 [-6.1, -1.0]	+7.2 [4.8, 10.9]	0.943	0.932	-0.3 [-2.5, 2.3]	+3.4 [2.1, 5.1]
Ours (full data, three-way)	0.938	0.866	+6.7 [4.0, 9.5]	9.5	0.891	0.873	+7.3 [4.3, 9.9]	-0.4 [-1.4, 0.6]	0.942	0.931	+8.9 [6.2, 12.1]	-1.0 [-2.4, 0.1]
second seed	0.941	0.883	+8.3 [5.6, 11.0]	6.7	0.887	0.875	+2.0 [-1.1, 4.5]	+0.4 [-1.1, 2.4]	0.945	0.931	+4.4 [2.2, 7.7]	+0.3 [-0.8, 1.7]
symmetric-view variant	0.941	0.868	+10.0 [7.0, 13.1]	9.2	0.899	0.881	+6.7 [3.9, 9.6]	-0.6 [-1.6, 0.7]	0.947	0.934	+8.5 [5.7, 12.0]	-0.9 [-2.2, -0.0]
same data, forged head only	0.949	0.896	+7.0 [4.7, 9.0]	7.4	0.914	0.883	+9.8 [7.4, 12.5]	-0.4 [-1.8, 0.9]	0.961	0.952	+4.2 [2.3, 6.5]	+0.2 [-0.9, 1.4]

Table S5: Celeb-DF-B: re-encoding separated from the preset on both errors, frame and video level. Video AUC: mean frame score per video. FA_c, FA_b: false alarms on re-encoded and beautified genuine frames at the threshold of 5% FPR on untreated genuine frames; $\Delta FA_{beauty} = FA_b - FA_c$ with a 95% video-cluster interval (1,000 draws, threshold re-estimated in each).

	frame AUC			video AUC			false alarms (%)		
Detector	untreated	re-encoded	beautified	untreated	re-encoded	beautified	FA_c	FA_b	ΔFA_{beauty} [95% CI]
Xception [2]	0.811	0.795	0.737	0.869	0.855	0.774	7.4	5.7	-1.7 [-3.0, -0.4]
EfficientNet-B4 [3]	0.811	0.798	0.762	0.870	0.849	0.816	6.1	7.8	+1.7 [+0.2, +3.6]
F3Net [4]	0.798	0.782	0.731	0.856	0.834	0.769	7.3	5.4	-1.9 [-3.5, -0.3]
SPSL [5]	0.781	0.769	0.727	0.849	0.832	0.775	5.8	3.4	-2.4 [-3.4, -1.6]
UCF [6]	0.841	0.829	0.777	0.897	0.883	0.818	5.6	5.7	+0.2 [-1.4, +1.7]
RECCE [7]	0.785	0.765	0.723	0.847	0.824	0.766	7.8	4.1	-3.6 [-5.0, -2.3]
CORE [8]	0.821	0.808	0.752	0.880	0.867	0.797	6.2	4.0	-2.2 [-3.3, -0.9]
SRM [9]	0.789	0.769	0.733	0.857	0.842	0.788	6.2	2.8	-3.4 [-4.3, -2.5]
SBI [14]	0.875	0.861	0.752	0.922	0.906	0.799	2.4	1.8	-0.6 [-1.6, +0.3]
UnivFD‡ [15]	0.653	0.637	0.624	0.702	0.681	0.658	3.1	7.6	+4.5 [+3.1, +6.7]
NPR‡ [16]	0.503	0.503	0.498	0.501	0.504	0.494	2.5	3.4	+0.9 [-0.3, +2.2]
Effort [10]	0.909	0.882	0.807	0.952	0.933	0.852	3.3	5.4	+2.1 [+0.2, +4.1]
Forensics Adapter [11]	0.909	0.894	0.828	0.961	0.949	0.888	7.8	7.9	+0.1 [-1.6, +2.3]
GenD [12]	0.924	0.898	0.807	0.971	0.953	0.854	6.3	9.9	+3.6 [+1.2, +6.6]
DFD-HR [13]	0.938	0.915	0.838	0.979	0.963	0.890	4.8	14.6	+9.8 [+6.5, +13.1]
CADDM [17]	0.886	0.850	0.762	0.965	0.933	0.832	3.9	13.8	+9.9 [+7.7, +12.1]
LAA-Net [18]	0.889	0.870	0.698	0.960	0.944	0.743	2.7	6.0	+3.3 [+1.6, +5.1]
OURS three-way, full data	0.938	0.918	0.866	0.979	0.966	0.920	4.6	9.5	+4.9 [+2.8, +7.5]
second seed	0.941	0.926	0.883	0.984	0.977	0.941	4.0	6.7	+2.7 [+1.3, +4.1]
symmetric-view variant	0.941	0.925	0.868	0.981	0.974	0.918	4.3	9.2	+4.9 [+2.7, +7.3]
OURS binary control	0.949	0.934	0.896	0.982	0.974	0.948	5.0	7.4	+2.3 [+1.2, +3.8]
OURS FF++ only, three-way	0.938	0.915	0.872	0.973	0.958	0.920	6.0	7.8	+1.8 [-0.1, +4.2]
OURS P0	0.927	0.901	0.860	0.961	0.940	0.909	6.7	5.1	-1.6 [-3.2, +0.1]
OURS P2 symmetric smoothing	0.920	0.900	0.866	0.961	0.942	0.916	6.7	5.5	-1.2 [-2.7, +0.5]
seed 2	0.933	0.914	0.888	0.963	0.947	0.928	6.6	6.2	-0.3 [-1.8, +1.3]

Table S6: Celeb-DF-B, paired differences of the frame AUC under the presets (AUC_b), our model minus each CLIP detector, with 95% intervals over identical video draws.

Model	Effort	Forensics Adapter	GenD	DFD-HR
OURS three-way, full data	+0.059 [+0.030, +0.089]	+0.038 [+0.013, +0.064]	+0.059 [+0.032, +0.086]	+0.028 [+0.005, +0.052]
second seed	+0.075 [+0.050, +0.103]	+0.055 [+0.031, +0.079]	+0.075 [+0.051, +0.104]	+0.045 [+0.022, +0.068]
symmetric-view variant	+0.061 [+0.032, +0.091]	+0.040 [+0.015, +0.065]	+0.061 [+0.035, +0.088]	+0.030 [+0.007, +0.054]
OURS binary control	+0.088 [+0.060, +0.119]	+0.068 [+0.042, +0.094]	+0.088 [+0.060, +0.118]	+0.058 [+0.034, +0.083]
OURS FF++ only, three-way	+0.065 [+0.041, +0.092]	+0.044 [+0.019, +0.068]	+0.065 [+0.041, +0.090]	+0.035 [+0.014, +0.055]
OURS P0 (FF++ only)	+0.053 [+0.026, +0.081]	+0.032 [+0.007, +0.058]	+0.053 [+0.028, +0.078]	+0.022 [+0.002, +0.042]
OURS P2 symmetric smoothing	+0.059 [+0.030, +0.088]	+0.038 [+0.011, +0.064]	+0.058 [+0.033, +0.088]	+0.028 [+0.006, +0.050]
seed 2	+0.081 [+0.055, +0.109]	+0.060 [+0.034, +0.085]	+0.081 [+0.056, +0.111]	+0.051 [+0.029, +0.074]

The data hold 232 videos per condition; 231 have at least one detected face in every condition except the beautified genuine one (230). Re-encoding alone lowers the frame AUC of every detector by at most 0.036, and the presets lower it further by 0.006–0.172 (Table S5); for the four CLIP detectors the loss beyond re-encoding is 0.066–0.091 and for OURS 0.052 and 0.043. On the false-alarm side, beyond re-encoding, the presets add 9.8 points for DFD-HR [6.5, 13.1], 9.9 for CADDM [7.7, 12.1] and 3.6 for GenD [1.2, 6.6], against 2.1 for Effort and 0.1 for Forensics Adapter; OURS adds 4.9 [2.8, 7.5] (second seed 2.7). At video level CADDM goes from 0.965 untreated to 0.933 re-encoded and 0.832 beautified; Libourel et al. report 0.91 to 0.76 for it on their own frames. Two confounds remain. The app re-frames treated videos to portrait with tinted bars, so face size and resampling differ between untreated and treated frames, and the re-encoded control reproduces the codec but not the re-framing; we did not build a face-size-matched subset. The subtraction assumes that the effects of re-encoding and preset add.

Table S7: The same videos, frames and per-video filter assignment with the filter applied to the face crop or to the full frame before the crop (frame AUC). A filter with a vignette is centred on the face in the first case. The paper reports the full-frame version.

	Celeb-DF-v2, 11 filters			DFD, 16 filters		
Detector	AUC	AUC_f crop	AUC_f full frame	AUC	AUC_f crop	AUC_f full frame
Xception [2]	0.718	0.673	0.693	0.854	0.823	0.834
EfficientNet-B4 [3]	0.750	0.698	0.725	0.810	0.777	0.781
F3Net [4]	0.708	0.663	0.688	0.781	0.769	0.774
SPSL [5]	0.703	0.664	0.683	0.829	0.834	0.837
UCF [6]	0.744	0.669	0.697	0.822	0.793	0.804
RECCE [7]	0.688	0.636	0.661	0.833	0.795	0.808
CORE [8]	0.715	0.669	0.693	0.827	0.813	0.821
SRM [9]	0.717	0.696	0.712	0.830	0.811	0.822
SBI [14]	0.817	0.729	0.776	0.829	0.803	0.810
UnivFD‡ [15]	0.662	0.610	0.622	0.840	0.793	0.801
NPR‡ [16]	0.522	0.502	0.497	0.473	0.541	0.516
Effort [10]	0.881	0.788	0.813	0.939	0.904	0.905
Forensics Adapter [11]	0.887	0.790	0.814	0.940	0.907	0.924
GenD [12]	0.891	0.818	0.856	0.934	0.913	0.925
DFD-HR [13]	0.879	0.781	0.824	0.943	0.921	0.928
Ours	0.891	0.849	0.873	0.942	0.927	0.931
Ours, FF++ only	0.898	0.797	0.851	0.943	0.925	0.932
binary control	0.914	0.863	0.883	0.961	0.949	0.952

S5 The decomposition of a commercial render

Algorithm S1: dense optical flow on locally normalised luminance gives the warp f between an original O and its render R, the photometric residual is taken in the geometry of O, and four single-operation components are formed, each with its measured strength and exact footprint. Applying the four in sequence reproduces R at 108 dB PSNR.

Algorithm S1 Single-operation components of a commercial render

Require: original O, render R, eye mask E, σ, noise floors τ_k

$f \leftarrow \text{DIS}\big(\text{LCN}(O), \text{LCN}(R)\big)$

$P \leftarrow R \circ (\text{id} - f^{-1}) - O$

$C_{\text{eye}} \leftarrow \mathcal{W}(O; E \odot f)$; $\quad C_{\text{contour}} \leftarrow \mathcal{W}(O; (1-E) \odot f)$

$C_{\text{tone}} \leftarrow O + G_\sigma * P$; $\quad C_{\text{texture}} \leftarrow O + P - G_\sigma * P$

for $k \in \{\text{eye}, \text{contour}, \text{tone}, \text{texture}\}$ **do**

    $q_k \leftarrow \text{Measure}_k(C_k, O)$; **if** $q_k \geq \tau_k$ **keep** $(C_k,\ e_k,\ M(C_k))$

end for

The loss is $\mathcal{L} = \text{BCE}(z_f, y_f) + \text{BCE}(z_b, y_b) + \mathbf{1}[y_b]\,\text{BCE}(\hat{s}, s) + \mathbf{1}[M \text{ exact}]\,\big(\text{BCE}(\hat{m}, M) + \text{Dice}(\hat{m}, M)\big)$, with $\hat{s}$ the operation-presence logits.

S6 Exposure arms on every source

Table S8: Which class receives beautified training views. FF++ only, forged head only, everything else identical; three seeds. Celeb-DF-B under the one rule of the paper.

Arm	Beautified training views	AUC_b (3 seeds)	$\Delta\text{MISS}_{\text{beauty}}$ (3 seeds)	FA_b
P0	no beautified views	0.860 / 0.883 / 0.883	+17.0 / +14.2 / +10.8	5.1 / 5.6 / 6.8
P1	beautified genuine faces only	0.844 / 0.851 / 0.864	+21.5 / +21.8 / +20.4	4.5 / 3.9 / 3.7
P2	beautified genuine and forged faces	0.879 / 0.893 / 0.877	+9.8 / +10.3 / +12.8	5.6 / 5.3 / 5.4
Paired differences, 95% video-cluster intervals (seeds 1 / 2 / 3)				
P1 − P0	$\Delta\text{MISS}_{\text{beauty}}$	+4.5 [1.5, 7.6] / +7.5 [4.4, 10.8] / +9.5 [6.7, 12.3]		
P2 − P0	$\Delta\text{MISS}_{\text{beauty}}$	-7.2 [-9.4, -5.0] / -4.0 [-6.1, -2.0] / +2.0 [0.1, 4.0]		
P2 − P1	$\Delta\text{MISS}_{\text{beauty}}$	-11.7 [-14.5, -8.8] / -11.5 [-14.5, -8.9] / -7.5 [-10.0, -5.0]		

Table S9: P0 none, P1 genuine only, P2 both; FF++ only, forged head only; values per seed. Effort scheme: the SVD-residual tuning of the released Effort code in place of LoRA; P1 was not trained under it. Frame AUC; ΔMISS at the 5%-FPR threshold of untreated genuine frames.

		Celeb-DF-B			Celeb-DF-v2 filters		DFD filters	
Tuning	Arm	AUC	AUC_b	$\Delta\text{MISS}_{\text{beauty}}$	AUC_f	ΔMISS	AUC_f	ΔMISS
LoRA	P0	0.927 / 0.937 / 0.944	0.860 / 0.883 / 0.883	+17.0 / +14.2 / +10.8	0.852 / 0.858 / 0.855	+7.1 / +4.4 / +8.3	0.924 / 0.928 / 0.932	+9.3 / +7.2 / +8.7
LoRA	P1	0.928 / 0.931 / 0.939	0.844 / 0.851 / 0.864	+21.5 / +21.8 / +20.4	0.798 / 0.820 / 0.838	+16.6 / +17.7 / +16.3	0.905 / 0.911 / 0.913	+11.2 / +9.3 / +9.2
LoRA	P2	0.929 / 0.933 / 0.935	0.879 / 0.893 / 0.877	+9.8 / +10.3 / +12.8	0.848 / 0.850 / 0.856	+0.3 / +3.0 / -0.9	0.929 / 0.929 / 0.927	+2.5 / +1.8 / -0.5
Effort scheme	P0	0.943 / 0.948	0.882 / 0.886	+10.1 / +10.7	0.876 / 0.886	+2.3 / -1.3	0.942 / 0.947	+8.7 / +7.4
Effort scheme	P2	0.938 / 0.943	0.889 / 0.890	+5.5 / +5.5	0.872 / 0.879	-7.6 / -8.4	0.948 / 0.953	+3.0 / +2.4

Bootstrap p-values for the nine exposure contrasts on Celeb-DF-B (P1−P0, P2−P0 and P2−P1 on $\Delta\text{MISS}_{\text{beauty}}$, three seeds, 1,000 identical video draws, $p = 2\min(\Pr[d \le 0], \Pr[d \ge 0])$ floored at 10^{-3}), Holm-adjusted over the nine: P1−P0 is $+4.5$, $+7.5$ and $+9.5$ points (adjusted $p = 0.009$ each); P2−P1 is -11.7, -11.5 and -7.5 (0.009 each); P2−P0 is -7.2 and -4.0 (0.009) and $+2.0$ on the third seed (0.044). The family-wise picture is the one the paper states: P1 is worse than P0 and P2 in every seed, and P2 against P0 changes sign across seeds. Paired video-cluster intervals for P2 minus P0 on the two filter sources: the miss rise falls by 5.0 to 9.9 points in nine of ten seed-by-source cells (intervals below zero; the tenth is -1.4 $[-3.3, 0.6]$), the false-alarm rise grows by 2.0 to 4.6 points in the same nine, and the frame AUC under the filters changes by -0.008 to $+0.006$ (intervals contain zero except for the Effort scheme on DFD, $+0.006$). For P1 minus P0 on Celeb-DF-v2 the miss rise grows by 13.3 and 8.0 points and the AUC under the filters falls by 0.038 and 0.017.

S7 The colour-filter probe, per filter

The probed model is the FF++-only three-way CLIP model ("OURS, FF++ only" in Table III), trained with random colour grading and 14 colour filters as beautified views; the training-time subspace below exists because those views were in its training. No published detector was probed.

Table S10: FF++ validation frames under the eleven held-out filters: cosine of the mean feature shift with the forged weight vector w, mean change of the forged logit on genuine and forged frames, and share of the shift energy inside the 16-dimensional subspace spanned by training-time beautifications.

Held-out filter	n	$\cos(w, \bar{\Delta})$	Δlogit genuine	Δlogit forged	energy in V_{16}
earlybird	88	0.20	+7.84	+1.10	0.86
mayfair	94	0.33	+5.64	+1.80	0.83
kelvin	92	0.04	+2.40	-0.71	0.90
clarendon	99	0.07	+1.90	+0.40	0.86
lofi	95	0.06	+1.86	+0.17	0.87
moon	89	0.06	+1.75	-0.10	0.75
perpetua	107	0.09	+0.99	+0.21	0.77
inkwell	95	0.09	+0.86	-0.25	0.63
reyes	104	0.02	+0.76	-0.27	0.89
slumber	101	0.00	+0.43	-0.28	0.87
brooklyn	95	0.00	+0.14	-0.04	0.88

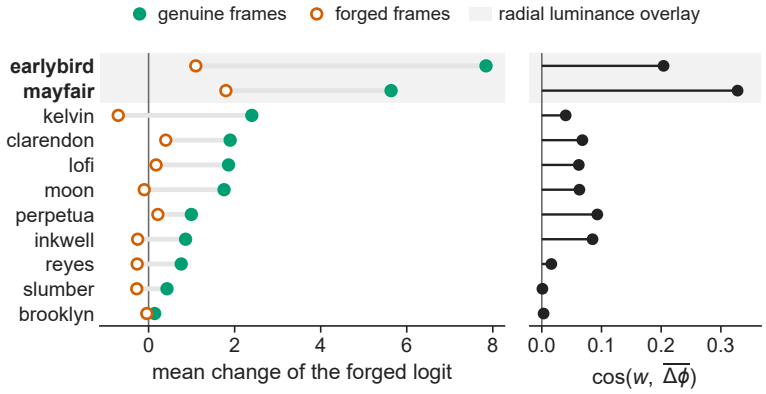


Figure S1: Mean change of the forged logit under each held-out filter on genuine (filled) and forged (open) frames (left), and cosine of the mean feature shift with w (right). Shaded rows: the two filters with a radial luminance overlay.

Table S11: Linear erasure. The feature shifts caused by training-time beautifications are decomposed by SVD on training frames, the first k directions are projected out, and a logistic forged head is refitted on training features. FA and MISS at the 5%-FPR threshold of clean genuine frames, pooled over the eleven held-out filters. The control is the refit without erasure ($k = 0$), not the trained head.

Forged head	AUC clean	AUC filtered	FA filtered	MISS filtered
trained head	0.987	0.953	21.5	5.9
refit, 0 directions erased	0.985	0.916	15.8	16.6
refit, 2 directions erased	0.933	0.907	18.5	15.1
refit, 8 directions erased	0.932	0.844	18.3	27.6
refit, 32 directions erased	0.919	0.804	19.2	32.1

The two leading directions hold 79% of the shift energy of training-time beautifications. Against the refit without erasure, removing them costs 0.052 clean AUC and raises the pooled false alarms under the filters from 15.8 to 18.5% (18.3 and 19.2% for 8 and 32 directions); per filter the false-alarm rate after erasure ranges from 17.5 to 27.5%. The directions along which beautification moves the feature carry forgery evidence for this model.

S8 Skin smoothing: the chain of pre-registered arms

All arms are FF++-only models with the forged head only; they share the architecture, data, steps and checkpoint rule of the exposure arms and differ only in which class receives beautified views and what those views contain. No image of any

retouching service is used. The scripted smoothing is a bilateral filter blended inside a feathered face hull, the same kind of edit as the service's (face-local and edge-preserving: high-frequency energy on flat skin 0.68 at the top of the training range against 0.60 for the service's level 90, background 1.00 against 0.96). Its strength range was set to cover the levels measured on the two sister services in RetouchingFFHQ while the test service's measurements were on disk.

Table S12: Commercial skin smoothing, every detector and arm: 2,210 FFHQ originals and their renders. Smoothing-90: share of renders scored above their own original (50 means no shift), medians of $P(\text{fake})$, and share above the threshold that accuses 5% of the originals. All renders: the four operations at three levels (26,519 images). NPR scores originals and renders at 1.0.

	Smoothing-90 renders			all renders	
Detector	above own original (%)	median $P(\text{fake})$ orig. $\rightarrow$ render	above 5%-FPR thr. (%)	above 5%-FPR thr. (%)	AUROC render vs orig.
Xception [2]	95.2	0.291 $\rightarrow$ 0.944	51.6	12.2	0.563
EfficientNet-B4 [3]	93.7	0.316 $\rightarrow$ 0.964	36.3	9.4	0.545
F3Net [4]	97.8	0.131 $\rightarrow$ 0.974	50.6	11.8	0.568
SPSL [5]	97.3	0.463 $\rightarrow$ 0.963	51.2	12.1	0.567
UCF [6]	95.4	0.031 $\rightarrow$ 0.984	41.4	11.2	0.575
RECCE [7]	96.5	0.120 $\rightarrow$ 0.944	36.8	9.5	0.558
CORE [8]	96.6	0.249 $\rightarrow$ 0.947	54.8	12.4	0.569
SRM [9]	99.3	0.461 $\rightarrow$ 0.668	62.8	13.9	0.570
SBI [14]	96.2	0.208 $\rightarrow$ 0.991	37.3	11.3	0.582
UnivFD‡ [15]	90.4	0.005 $\rightarrow$ 0.019	16.6	7.3	0.526
NPR‡ [16]	n/a (saturated)	n/a	n/a	n/a	n/a
Effort [10]	95.6	0.319 $\rightarrow$ 0.595	43.2	14.8	0.599
Forensics Adapter [11]	93.3	0.058 $\rightarrow$ 0.354	39.7	10.2	0.541
GenD [12]	98.5	0.228 $\rightarrow$ 0.921	68.7	16.8	0.596
DFD-HR [13]	99.0	0.210 $\rightarrow$ 0.711	64.9	15.7	0.554
CADDM [17]	60.9	0.538 $\rightarrow$ 0.745	6.3	5.6	0.523
LAA-Net [18]	89.2	0.234 $\rightarrow$ 0.969	17.2	7.6	0.578
OURS, three-way, full data	32.5	0.002 $\rightarrow$ 0.001	8.6	4.7	0.371
OURS, symmetric-view variant	30.0	0.003 $\rightarrow$ 0.001	8.1	4.8	0.375
OURS, binary control	70.5	0.000 $\rightarrow$ 0.001	9.1	5.2	0.490
OURS, FF++ only, exposure arms (seed 1)					
P0	98.5	0.049 $\rightarrow$ 0.968	54.2	13.5	0.576
P2	84.1	0.009 $\rightarrow$ 0.116	11.7	5.9	0.528
P2 no smoothing	98.6	0.017 $\rightarrow$ 0.988	55.3	12.7	0.553
P2 smoothing only	27.3	0.053 $\rightarrow$ 0.020	2.3	3.7	0.449
P1	43.5	0.004 $\rightarrow$ 0.004	3.7	5.3	0.500
P1 smoothing only	2.6	0.074 $\rightarrow$ 0.000	0.7	4.9	0.437
P2 symmetric smoothing	53.7	0.017 $\rightarrow$ 0.022	5.9	4.9	0.488
OURS pipeline, FF++ only, Effort's tuning scheme (seed 1)					
Effort scheme, no beautified views	99.1	0.028 $\rightarrow$ 0.963	56.9	13.3	0.587
Effort scheme, symmetric smoothing	88.4	0.029 $\rightarrow$ 0.095	14.5	7.2	0.541
released Effort fine-tuned, no beautified views	98.5	0.091 $\rightarrow$ 0.306	55.3	14.4	0.564
released Effort fine-tuned, symmetric smoothing	96.6	0.080 $\rightarrow$ 0.174	39.9	11.5	0.553

Table S13: Paired shift of the forgery logit, Smoothing-90 render minus its own original (2,210 pairs): counts of pairs moving up and down, median shift with a 95% bootstrap interval over originals (1,000 draws), and the two-sided sign test with ties dropped. Probabilities are clipped to $[10^{-7}, 1-10^{-7}]$ before the logit. NPR is saturated and omitted.

Model	up / down / tie	median Δlogit [95% CI]	sign test p
Xception [2]	2105 / 105 / 0	+3.32 [+3.17, +3.45]	$< 10^{-10}$
EfficientNet-B4 [3]	2070 / 140 / 0	+3.48 [+3.33, +3.57]	$< 10^{-10}$
F3Net [4]	2161 / 49 / 0	+4.92 [+4.74, +5.08]	$< 10^{-10}$
SPSL [5]	2151 / 59 / 0	+3.05 [+2.95, +3.12]	$< 10^{-10}$
UCF [6]	2108 / 102 / 0	+5.47 [+5.20, +5.73]	$< 10^{-10}$
RECCE [7]	2133 / 77 / 0	+3.82 [+3.70, +3.96]	$< 10^{-10}$
CORE [8]	2135 / 75 / 0	+3.90 [+3.74, +4.03]	$< 10^{-10}$
SRM [9]	2194 / 16 / 0	+0.68 [+0.65, +0.71]	$< 10^{-10}$
SBI [14]	2127 / 83 / 0	+4.88 [+4.72, +5.04]	$< 10^{-10}$
UnivFD‡ [15]	1998 / 212 / 0	+1.22 [+1.17, +1.28]	$< 10^{-10}$
Effort [10]	2113 / 97 / 0	+0.96 [+0.90, +1.02]	$< 10^{-10}$
Forensics Adapter [11]	2063 / 147 / 0	+1.72 [+1.59, +1.87]	$< 10^{-10}$
GenD [12]	2176 / 34 / 0	+3.27 [+3.19, +3.35]	$< 10^{-10}$
DFD-HR [13]	2189 / 21 / 0	+2.06 [+1.93, +2.15]	$< 10^{-10}$
CADDM [17]	1345 / 865 / 0	+0.67 [+0.53, +0.81]	$< 10^{-10}$
LAA-Net [18]	1971 / 239 / 0	+3.60 [+3.36, +3.79]	$< 10^{-10}$
Ours three-way, full data	718 / 1490 / 2	-1.20 [-1.28, -1.12]	$< 10^{-10}$
second seed	913 / 1296 / 1	-0.53 [-0.60, -0.41]	$< 10^{-10}$
symmetric-view variant	662 / 1545 / 3	-1.09 [-1.19, -1.00]	$< 10^{-10}$
Ours binary control	1557 / 642 / 11	+0.41 [+0.38, +0.44]	$< 10^{-10}$
P0	2176 / 34 / 0	+5.17 [+5.05, +5.38]	$< 10^{-10}$
P2	1859 / 347 / 4	+1.81 [+1.68, +1.97]	$< 10^{-10}$
P1	962 / 1245 / 3	-0.22 [-0.28, -0.16]	1.8×10^{-9}
P1-smooth	57 / 2153 / 0	-5.00 [-5.16, -4.88]	$< 10^{-10}$
P2-smooth	604 / 1602 / 4	-0.91 [-0.97, -0.84]	$< 10^{-10}$
P2-sym	1187 / 1017 / 6	+0.09 [+0.03, +0.12]	3.2×10^{-4}
P2-sym, seed 2	1144 / 1059 / 7	+0.03 [-0.00, +0.09]	0.07

The published detectors and the no-exposure arm move the logit of a smoothed genuine face up by 0.7 to 5.5. The full-data three-way models and the genuine-only or form-asymmetric arms move it down, which is the evasion direction; the binary control moves it up by 0.41. Only P2-sym is close to no shift, with a small upward median on its first seed ($p = 3.2 \times 10^{-4}$) and none detectable on its second ($p = 0.07$). A 25–75% band on the share of pairs, the bar used for P2-sym and the variant, is wide: it admits Ours's 32.5%, which the sign test shows is a shift.

The no-exposure arm reproduces the published detectors. P0 accuses 54.2, 56.1 and 43.5% (three seeds) of the Smoothing-90 renders at the 5%-FPR threshold of the service's originals, scores the render above its own original in 97.0–99.0% of pairs, and moves the median $P(\text{fake})$ from 0.05 to 0.97 (seed 1). P0 is at the 5% floor on the other nine operation-by-level cells of the service (whitening 4.5–7.5%, face slimming 4.5–6.1%, eye enlarging 4.8–5.9%) and dose-dependent on smoothing (12.6–17.0, 29.5–40.8 and 43.5–56.1% at levels 30, 60 and 90). P0 trains with label-symmetric Gaussian blur, down-up scaling and JPEG on half of its triplets, so plain blur or re-compression in training does not prevent the accusation. With beautified views on genuine faces only (P1) the Smoothing-90 rate falls to 3.7, 7.1 and 4.6%; with views on both classes (P2) to 11.7, 7.1 and 6.9% (paired differences against P0 of -37 to -51 points, every interval below zero). The rate holds under every threshold rule tried for these arms (1% and 10% FPR, a fixed 0.5, thresholds set on Celeb-DF-v2 or DFD genuine frames). P2 still scores the render above its original in 72–84% of pairs (AUROC 0.63–0.70). Both arms pay for it on untreated Celeb-DF-v2: frame AUC -0.012 to -0.021 for P2 and -0.025 to -0.030 for P1 against P0 on every seed, and 5–8 points of miss rate at 5% FPR. This first analysis was post hoc, and its per-epoch curve for the both-class arm (27% at epoch 2, 9–12% from epoch 5, with checkpoints chosen on FF++ validation AUC by margins of 0.0003–0.002) is disclosed for that reason. Inside Effort's SVD-residual scheme the P2 views leave 30.0 and 38.9% (P0 under that scheme 56.9 and 63.2%), and two epochs of fine-tuning the released Effort checkpoint with them leave 52.7% against 55.3% without them.

No other view substitutes for smoothing. Declared reading: if the arm stays at 30% or more on both seeds, the effect is exposure to smoothing itself. P2-nosmooth removes smoothing from every beautified view of P2 and keeps colour grades, the fourteen colour filters, whitening, eye and jaw edits, and the shared degradation. Smoothing-90 is then accused at 55.3% [51.9, 58.6] and 67.1% (two seeds), against P0 of the same seed $+1.2$ [-2.2, 4.0] and $+11.1$ [7.0, 14.8], and the render is ranked above its original in 98.6 and 99.4% of pairs. The genuine-only version without smoothing gives 55.9% ($+1.8$ [-1.8, 5.2] against P0). The declared reading holds within the views we trained. These arms still lose untreated

Table S14: The trade-off table of the paper (Table V) with every column; values per seed (1 / 2 / 3 where trained). Arm names in the paper: P2 no smoothing = P2-nosmooth, P2 smoothing only = P2-smooth, P1 smoothing only = P1-smooth, P2 symmetric smoothing = P2-sym. Untreated video AUC; Celeb-DF-B frame AUC; DF40 reenactment video AUC.

	commercial Smoothing-90		smoothed Celeb-DF-v2 (strong bil.)		untreated video AUC		Celeb-DF-B (frame)		DF40 FR video AUC	
Arm	accused (%)	above own orig. (%)	miss rise	frame AUC, both smoothed	Celeb-DF-v2	DFD	AUC_b	$\Delta MISS_{beauty}$	FF++ dom.	Celeb-DF dom.
P0	54.2 / 56.1 / 43.5	98.5 / 99.0 / 97.0	-16.0 / -11.8 / +4.3	0.747 / 0.736 / 0.736	0.938 / 0.938 / 0.940	0.981 / 0.979 / 0.978	0.860 / 0.883 / 0.883	+17.0 / +14.2 / +10.8	0.906 / 0.877 / 0.873	0.692 / 0.720 / 0.633
P2	11.7 / 7.1 / 6.9	84.1 / 72.4 / 75.5	-26.5 / -13.3 / –	0.740 / 0.746 / –	0.917 / 0.920 / 0.935	0.973 / 0.975 / 0.972	0.879 / 0.893 / 0.877	+9.8 / +10.3 / +12.8	0.809 / 0.785 / –	0.644 / 0.643 / –
P2 no smoothing	55.3 / 67.1	98.6 / 99.4	-28.9 / -33.1	0.732 / 0.695	0.913 / 0.924	0.980 / 0.973	0.879 / 0.873	+12.9 / +12.9	0.858 / –	0.705 / –
P2 smoothing only	2.3 / 5.3	27.3 / 51.7	-48.2 / -44.6	0.785 / 0.734	0.904 / 0.926	0.967 / 0.966	0.865 / 0.870	+10.2 / +13.7	0.779 / 0.833	0.593 / 0.629
P1	3.7 / 7.1 / 4.6	43.5 / 67.5 / 64.3	+51.2 / – / –	0.587 / – / –	0.900 / 0.910 / 0.923	0.978 / 0.975 / 0.971	0.844 / 0.851 / 0.864	+21.5 / +21.8 / +20.4	0.807 / – / –	0.569 / – / –
P1 smoothing only	0.7 / 1.0 / 0.9	2.6 / 6.5 / 5.4	+56.4 / +60.0 / +62.9	0.643 / 0.579 / 0.596	0.922 / 0.929 / 0.939	0.974 / 0.977 / 0.964	0.887 / 0.896 / 0.893	+8.6 / +8.5 / +7.3	0.871 / 0.869 / 0.900	0.673 / 0.689 / 0.679
P2 symmetric smoothing	5.9 / 6.1	53.7 / 51.8	+14.6 / +17.5	0.767 / 0.796	0.919 / 0.934	0.972 / 0.968	0.866 / 0.888	+14.2 / +10.4	0.871 / 0.796	0.670 / 0.646
Effort scheme, no beautified views	56.9 / 63.2	99.1 / 99.1	-32.2 / –	0.719 / –	0.944 / 0.954	0.987 / 0.991	0.882 / 0.886	+10.1 / +10.7	– / –	– / –
Effort scheme, symmetric smoothing	14.5	88.4	+24.9	0.757	0.935	0.986	0.890	+10.6	0.822	0.614
released Effort fine-tuned, no beautified views	55.3	98.5	-39.9	0.754	0.926	0.976	0.805	+5.8	–	–
released Effort fine-tuned, symmetric smoothing	39.9	96.6	-19.0	0.740	0.921	0.974	0.791	+7.5	0.805	0.705
OURS, three-way, full data	8.6 / 12.5	32.5 / 41.3	-44.3 / –	0.688 / –	0.947 / 0.946	0.978 / 0.981	0.866 / 0.883	+6.7 / +8.3	0.718 / 0.780	0.502 / 0.559
OURS, symmetric-view variant	8.1	30.0	+34.1	0.796	0.954	0.983	0.868	+10.0	0.738	0.539
OURS, binary control	9.1	70.5	-15.0	0.754	0.956	0.987	0.896	+7.0	0.842	0.647

Table S15: Smoothed forgeries and smoothed genuine frames. Celeb-DF-v2 test frames (every fourth of the 32 per video), untreated and under a whole-crop bilateral filter at three strengths, a 5×5 median filter and a 50% blend with a Gaussian blur; the last two forms were shown to no arm. Threshold at 5% FPR on untreated genuine frames. Miss rise: change of the miss rate on forgeries after smoothing, with 95% video-cluster intervals. Last column: frame AUC with both classes under the strong bilateral filter.

		miss rise (points, 95% CI)					false alarms, smoothed genuine (%)					
Arm	miss untr.	bil. light	bil. mid	bil. strong	median 5	Gauss. blend	bil. light	bil. mid	bil. strong	median 5	Gauss. blend	AUC both smoothed (strong)
P0	42.2	+27.0 [+24.4, +29.7]	+5.1 [-0.3, +10.5]	-16.0 [-21.5, -10.2]	+19.2 [+17.1, +21.2]	+3.0 [+1.4, +5.1]	0.8	16.8	39.6	1.5	5.5	0.747
seed 2	37.7	+26.7 [+24.1, +29.9]	+6.1 [+2.2, +11.8]	-11.8 [-15.5, -5.4]	+18.0 [+16.0, +20.4]	+8.1 [+6.6, +9.9]	1.6	18.2	42.9	1.9	4.4	0.736
seed 3	36.7	+24.0 [+21.4, +26.9]	+14.3 [+8.8, +17.9]	+4.3 [-2.4, +11.0]	+12.6 [+10.6, +14.6]	+5.6 [+4.2, +6.9]	2.3	11.3	25.2	3.0	4.9	0.736
P2	46.8	+16.5 [+14.2, +19.4]	-17.4 [-20.7, -13.1]	-26.5 [-29.7, -21.9]	+13.2 [+11.7, +15.8]	+7.0 [+6.0, +8.9]	3.0	31.8	49.1	2.5	3.3	0.740
P2 no smoothing	46.7	+19.8 [+16.8, +22.0]	-12.2 [-16.3, -6.5]	-28.9 [-32.8, -24.9]	+8.1 [+6.4, +9.8]	+2.2 [+0.7, +3.5]	2.3	25.6	54.4	3.9	4.7	0.732
P2 smoothing only	49.5	-34.3 [-37.6, -30.4]	-47.3 [-51.6, -40.2]	-48.2 [-53.1, -40.8]	+17.2 [+14.7, +20.2]	+10.9 [+9.2, +13.6]	47.8	82.3	88.3	1.7	2.4	0.785
P1	48.8	+45.9 [+39.7, +51.1]	+51.1 [+43.6, +57.2]	+51.2 [+43.7, +57.5]	+29.3 [+25.1, +32.4]	+6.4 [+4.8, +7.6]	0.0	0.0	0.0	0.7	4.7	0.587
P1 smoothing only	43.6	+56.4 [+49.9, +63.7]	+56.4 [+50.0, +63.9]	+56.4 [+50.0, +63.9]	+40.0 [+35.8, +44.4]	+10.8 [+9.1, +12.6]	0.0	0.0	0.0	0.4	4.0	0.643
seed 2	40.0	+58.7 [+52.3, +65.1]	+60.0 [+53.7, +66.7]	+60.0 [+53.7, +66.7]	+38.6 [+34.6, +42.6]	+13.9 [+12.3, +16.1]	0.0	0.0	0.0	0.7	3.1	0.579
seed 3	37.1	+59.3 [+53.9, +64.5]	+62.8 [+57.4, +68.6]	+62.9 [+57.5, +68.7]	+38.0 [+33.8, +41.7]	+13.5 [+11.9, +15.3]	0.0	0.0	0.0	0.5	2.7	0.596
P2 symmetric smoothing	46.2	+14.8 [+12.4, +17.0]	+14.5 [+11.4, +16.9]	+14.6 [+11.3, +17.6]	+7.6 [+5.4, +9.3]	+9.1 [+7.7, +10.6]	2.4	5.0	6.1	4.5	3.6	0.767
seed 2	42.0	+19.0 [+16.7, +21.5]	+20.6 [+17.3, +23.6]	+17.5 [+14.7, +20.5]	+9.0 [+7.2, +10.6]	+12.7 [+11.0, +14.3]	2.2	4.2	5.7	4.5	3.0	0.796
Effort scheme, no beautified views	39.7	+19.0 [+16.2, +21.7]	-11.3 [-15.8, -6.3]	-32.2 [-36.2, -27.5]	+11.2 [+8.4, +13.5]	-2.1 [-3.9, +0.1]	3.6	35.4	75.1	4.8	7.6	0.719
Effort scheme, symmetric smoothing	41.7	+22.4 [+20.0, +25.1]	+25.0 [+21.3, +28.5]	+24.9 [+19.3, +28.5]	+5.8 [+3.4, +7.9]	+7.2 [+5.8, +8.7]	2.0	4.2	6.1	7.3	4.3	0.757
released Effort fine-tuned, no beautified views	42.9	+17.2 [+12.3, +20.0]	-21.6 [-26.0, -14.6]	-39.9 [-44.7, -34.7]	+9.8 [+6.3, +12.2]	+1.1 [-3.2, +4.3]	4.2	39.2	81.2	7.7	6.4	0.754
released Effort fine-tuned, symmetric smoothing	43.7	+23.0 [+18.3, +26.4]	+6.6 [-1.3, +13.5]	-19.0 [-26.4, -10.1]	+10.0 [+5.9, +12.6]	-1.2 [-6.6, +2.9]	2.2	15.7	43.0	7.0	7.3	0.740
Ours, three-way, full data	47.0	-25.1 [-28.6, -21.8]	-42.3 [-46.3, -36.6]	-44.3 [-49.1, -38.3]	+4.5 [+1.7, +6.9]	+5.4 [+3.7, +6.6]	34.9	79.8	90.1	8.1	5.2	0.688
Ours, symmetric-view variant	43.5	+28.1 [+25.2, +30.6]	+33.8 [+29.9, +36.8]	+34.1 [+30.2, +37.1]	+12.8 [+11.0, +15.1]	+3.8 [+2.2, +5.1]	1.0	1.6	1.7	2.4	5.3	0.796
Ours, binary control	36.5	+0.7 [-3.2, +7.5]	-15.4 [-21.1, -2.5]	-15.0 [-20.8, -0.7]	+11.9 [+9.9, +14.0]	+6.2 [+5.1, +8.0]	9.9	35.5	45.6	4.7	5.4	0.754

Celeb-DF-v2 video AUC against P0 (0.913 and 0.924 against 0.938 and 0.938), so that cost is not caused by smoothing alone.

Smoothing as the only view, on both classes (P2-smooth). Bars, against P0 of the same seed, on both seeds: (a) Smoothing-90 at most 15% at the 5%-FPR threshold of the originals; (b) untreated Celeb-DF-v2 and DFD video AUC not lower by more than 0.010; (c) Celeb-DF-B frame AUC under the presets not lower by more than 0.010. Smoothing-90 falls to 2.3% [1.7, 2.9] and 5.3% [4.4, 7.0], -51.9 [-56.1, -48.3] and -50.7 [-55.3, -46.5] against P0: (a) is met on both seeds. Bar (b) is not met: untreated Celeb-DF-v2 video AUC is 0.904 and 0.926 against 0.938 and 0.938, untreated DFD 0.967 and 0.966 against 0.981 and 0.979. Bar (c) is 0.865 and 0.870 against 0.860 and 0.883, and the second seed is 0.013 lower, so it is not met. The arm was later found to carry the class-asymmetric form shortcut described below: it calls 47.8, 82.3 and 88.3% of whole-crop-smoothed genuine frames forged, and it ranks only 27.3% of the commercial renders above their originals (seed 1), which may come from the same cue: the commercial smoothing is face-local, the form shown on genuine faces in training. Its low accusation is therefore not evidence that smoothing views suffice; the class-symmetric arm below is.

The genuine-only recipe is an evasion route. P1-smooth lowers the accusation to 0.7, 1.0 and 0.9% (three seeds), ranks the render above its original in 2.6, 6.5 and 5.4% of pairs (AUROC 0.095, seed 1), and has the best Celeb-DF-B numbers of the LoRA arms (frame AUC under the presets 0.887, 0.896 and 0.893 against P0 0.860, 0.883 and 0.883). A check written before any smoothed forgery was scored asks what such a model does to a forgery that a forger smooths (Table S15). Its miss rate goes from 43.6, 40.0 and 37.1% on untreated forgeries to 96.4–100% under the bilateral forms, 75.1–83.6% under the median filter and 50.6–54.4% under the Gaussian blend (miss rises of $+56$ to $+63$, $+38$ to $+40$ and $+11$ to $+14$ points; lower interval bounds of $+49.9$ or more on the bilateral forms), with the forged score falling in 100% of frames and false alarms on smoothed genuine frames at 0.0%. P0 of the third seed rises by $+24.0$, $+14.3$, $+4.3$, $+12.6$ and $+5.6$. The arm with all beautified views on genuine faces only behaves the same ($+46$ to $+51$ on the bilateral forms). The shortcut "smoothed means genuine" removed the accusation and carries over to smoothing forms the views did not contain. The check fails on all three seeds.

The both-class arms carry a shortcut of the opposite sign. A whole-crop bilateral filter on a *genuine* frame is called forged by P2-smooth in 47.8, 82.3 and 88.3% of frames at the three strengths, by OURS in 34.9, 79.8 and 90.1%, by P2 in 3.0, 31.8 and 49.1%, against P0 at 0.8, 16.8 and 39.6% and the binary control at 9.9, 35.5 and 45.6%. In the training code genuine items receive the face-local scripted smoothing and pre-rendered smoothing frames, while forged items without landmarks receive only the whole-crop bilateral filter, so the form of the smoothing predicts the class. This class-asymmetric augmentation affects every both-class arm and OURS.

A class-symmetric view (P2-sym). A smoothed view is drawn the same way wherever landmarks exist: with probability 0.5 the face-local scripted smoothing, otherwise the whole-crop bilateral filter at one of three strengths; forged frames without landmarks receive the whole-crop form; no pre-rendered smoothing frames. Both forms occur in both classes, in proportions that are not identical (genuine 0.5 / 0.5; forged with landmarks 0.33 / 0.67; forged without landmarks 0 / 1). Bars: (a) Smoothing-90 at most 15% at the 5%-FPR rule and the render ranked above its original in 25–75% of pairs; (b) false alarms on smoothed genuine Celeb-DF-v2 frames at most 15% for every form; (c) a miss rise on smoothed forgeries at most 5 points above P0's for every form; two seeds. Bars (a) and (b) are met on both seeds: Smoothing-90 5.9 and 6.1%, the render above its original in 53.7 and 51.8% of pairs, false alarms on smoothed genuine frames 2.4, 5.0, 6.1, 4.5 and 3.6% (seed 1) and 2.2, 4.2, 5.7, 4.5 and 3.0% (seed 2) across the five forms. Bar (c) is not met: the miss rises are $+14.8$, $+14.5$, $+14.6$, $+7.6$ and $+9.1$ (seed 1) and $+19.0$, $+20.6$, $+17.5$, $+9.0$ and $+12.7$ (seed 2), against P0's $+27.0$, $+5.1$, -16.0, $+19.2$ and $+3.0$ and $+26.7$, $+6.1$, -11.8, $+18.0$ and $+8.1$; the arm exceeds the reference by more than 5 points at the mid and strong bilateral strengths on both seeds and under the Gaussian blend on the first (by 4.6 on the second). Part of the reason is the reference: P0's miss rate falls under strong smoothing because P0 accuses everything smoothed (false alarms of 39.6 and 42.9% on smoothed genuine frames at that strength). The arm's own misses still rise by 8–21 points. Read without a threshold, with both classes smoothed, its frame AUC is 0.835, 0.796, 0.767, 0.829 and 0.850 (seed 1) and 0.849, 0.809, 0.796, 0.853 and 0.865 (seed 2) against P0's 0.848, 0.776, 0.747, 0.852 and 0.868 and 0.852, 0.781, 0.736, 0.855 and 0.871: above P0 at the two heavier bilateral strengths, lower by at most 0.023 elsewhere. On untreated data its Celeb-DF-v2 video AUC is 0.919 and 0.934 against P0's 0.938 and 0.938 and its DFD video AUC 0.972 and 0.968 against 0.981 and 0.979. On Celeb-DF-B its frame AUC under the presets is 0.866 and 0.888 against 0.860 and 0.883, its beautification-specific miss rise $+14.2$ and $+10.4$ against $+17.0$ and $+14.2$. On DF40 its video AUC is 0.968 and 0.950 on face swap and 0.871 and 0.796 on reenactment in the FF++ domain, 0.913 and 0.935 and 0.670 and 0.646 in the Celeb-DF domain.

Under Effort's tuning scheme. Declared: the same view, FF++ only, in Effort's scheme from CLIP and as a two-epoch fine-tune of the released Effort checkpoint (learning rate 2×10^{-5}), one seed, bars (a)–(c) against each scheme's own reference; both pass: scheme-independent; one passes: reported per scheme; neither: the LoRA result stands alone.

Effort scheme from CLIP: (b) false alarms on smoothed genuine frames 2.0, 4.2, 6.1, 7.3 and 4.3% (reference 3.6, 35.4, 75.1, 4.8 and 7.6%), met. (a) Smoothing-90 14.5% (reference 56.9%), but the render above its own original in 88.4% of pairs, not met. (c) miss rise $+22.4$, $+25.0$, $+24.9$, $+5.8$ and $+7.2$ against $+19.0$, -11.3, -32.2, $+11.2$ and -2.1, not met. Released Effort fine-tuned for two epochs: Smoothing-90 39.9%, above own original 96.6% (reference 55.3% and 98.5%), (a) not met. Neither passes; the LoRA result stands alone, and the view removes the accusation of the commercial service only under LoRA.

The full-data model with the symmetric view. Declared: the recipe of OURS with the symmetric view (the scripted smoothing of a genuine view is, with probability 0.5, the whole-crop bilateral filter; pre-rendered smoothing frames dropped; everything else unchanged), seed 20260929, checkpoint by OURS's rule. It replaces OURS if (i) whole-crop-smoothed genuine Celeb-DF-v2 false alarms are at most 15% at every strength, (ii) Smoothing-90 is at most 15% with the render above its original in 25–75% of pairs, (iii) untreated Celeb-DF-v2 and DFD video AUC are not lower by more than 0.010, (iv) the explanation flip rate on the three part-forgery editors is not lower by more than 1.0 point, (v) Celeb-DF-B frame AUC under the presets is not lower by more than 0.010. Result: (i) 1.0, 1.6 and 1.7% (was 34.9, 79.8 and 90.1%), met. (ii) 8.1% and 30.0% (was 8.6% and 32.5%), met. (iii) Celeb-DF-v2 0.954 (0.947) and DFD 0.983 (0.978), met. (iv) flip rate on donor transplant, SD-1.5 and SDXL inpainting 92.6, 92.2 and 91.9% (94.0, 93.6 and 93.5%), lower by 1.4, 1.4 and 1.6 points: not met. The ceiling fell by the same amount (92.6, 92.1 and 92.0%; was 94.1, 93.7 and 93.6%), so flip minus ceiling is 0.0, $+0.1$ and -0.1 points against -0.1, -0.1 and -0.1, and IoU_7 is unchanged (0.836, 0.843 and 0.833). The bar was absolute; a bar relative to the ceiling, which we did not declare, would have been met. (v) 0.868 (0.866), with a beautification-specific miss rise of $+10.0$ ($+6.7$) and false alarms on beautified genuine frames of 9.2% (9.5%). Also: validation selection score 0.871 (0.862); untouched originals of the service sent to the beautified class 13.9% (20.8%); frame AUC with both classes smoothed 0.878, 0.831, 0.796, 0.880 and 0.878 (0.803, 0.740, 0.688, 0.819 and 0.855); miss rise on smoothed forgeries $+28.1$, $+33.8$, $+34.1$, $+12.8$ and $+3.8$ (miss rate 43.5% untreated, 77.6% under the strongest bilateral form), where OURS's -25 to -44 came from accusing everything smoothed. By the rule, (iv) fails, so OURS stays and the retrained model is reported next to it as the symmetric-view variant. On DF40 the variant reaches 0.952 and 0.909 on face swap and 0.738 and 0.539 on reenactment (OURS 0.952, 0.905, 0.718 and 0.502).

S9 DF40: every detector and arm

Table S16: DF40 held-out, untreated. Video AUC (mean frame score per video), mean over methods, with 95% bootstrap intervals over videos. Real videos are the FF++ and Celeb-DF-v2 test reals; the 718 MobileSwap videos from FF++ training identities were removed; 140 fake videos per method in the Celeb-DF domain. FS: nine face-swap methods; FR: twelve reenactment methods. CADDM and the rows from P0 downwards were scored after the first table had been read.

	FF++ domain		Celeb-DF domain	
Detector	FS	FR	FS	FR
Xception [2]	0.854 [0.839, 0.869]	0.859 [0.846, 0.872]	0.670 [0.653, 0.688]	0.691 [0.674, 0.706]
EfficientNet-B4 [3]	0.812 [0.796, 0.828]	0.818 [0.804, 0.831]	0.724 [0.706, 0.743]	0.712 [0.697, 0.729]
F3Net [4]	0.826 [0.809, 0.841]	0.797 [0.782, 0.811]	0.697 [0.679, 0.715]	0.671 [0.654, 0.686]
SPSL [5]	0.766 [0.748, 0.783]	0.754 [0.737, 0.770]	0.717 [0.699, 0.739]	0.704 [0.688, 0.721]
UCF [6]	0.843 [0.828, 0.857]	0.825 [0.811, 0.839]	0.716 [0.698, 0.733]	0.704 [0.689, 0.720]
RECCE [7]	0.860 [0.846, 0.874]	0.850 [0.837, 0.863]	0.736 [0.719, 0.753]	0.714 [0.698, 0.729]
CORE [8]	0.850 [0.835, 0.865]	0.858 [0.846, 0.870]	0.656 [0.638, 0.674]	0.676 [0.659, 0.693]
SRM [9]	0.836 [0.820, 0.852]	0.830 [0.816, 0.842]	0.732 [0.715, 0.750]	0.740 [0.725, 0.756]
Effort [10]	0.944 [0.935, 0.953]	0.808 [0.793, 0.822]	0.935 [0.926, 0.944]	0.710 [0.692, 0.726]
Forensics Adapter [11]	0.948 [0.940, 0.956]	0.828 [0.814, 0.841]	0.897 [0.885, 0.908]	0.665 [0.649, 0.681]
GenD [12]	0.966 [0.959, 0.973]	0.935 [0.927, 0.943]	0.947 [0.940, 0.955]	0.828 [0.815, 0.840]
DFD-HR [13]	0.982 [0.977, 0.986]	0.932 [0.924, 0.940]	0.940 [0.931, 0.948]	0.698 [0.681, 0.714]
CADDM [17]	0.869 [0.856, 0.882]	0.794 [0.779, 0.809]	0.788 [0.772, 0.804]	0.541 [0.523, 0.560]
OURS, three-way, full data	0.952 [0.943, 0.960]	0.718 [0.701, 0.733]	0.905 [0.894, 0.915]	0.502 [0.484, 0.519]
second seed	0.960 [0.952, 0.967]	0.780 [0.764, 0.793]	0.928 [0.918, 0.937]	0.559 [0.540, 0.577]
OURS, symmetric-view variant	0.952 [0.944, 0.959]	0.738 [0.722, 0.752]	0.909 [0.898, 0.919]	0.539 [0.521, 0.556]
OURS, binary control	0.969 [0.962, 0.975]	0.842 [0.828, 0.854]	0.926 [0.916, 0.936]	0.647 [0.630, 0.663]
OURS, FF++ only, three-way	0.962 [0.954, 0.969]	0.809 [0.794, 0.823]	0.933 [0.923, 0.942]	0.637 [0.621, 0.653]
P0	0.981 [0.976, 0.985]	0.906 [0.896, 0.915]	0.925 [0.916, 0.935]	0.692 [0.675, 0.707]
P0, seed 2	0.970 [0.964, 0.975]	0.877 [0.865, 0.887]	0.939 [0.930, 0.948]	0.720 [0.704, 0.734]
P0, seed 3	0.979 [0.974, 0.983]	0.873 [0.861, 0.884]	0.931 [0.923, 0.940]	0.633 [0.616, 0.648]
P2	0.942 [0.933, 0.950]	0.809 [0.794, 0.823]	0.907 [0.895, 0.917]	0.644 [0.627, 0.659]
P2, seed 2	0.944 [0.935, 0.952]	0.785 [0.770, 0.799]	0.901 [0.890, 0.912]	0.643 [0.627, 0.658]
P2 no smoothing	0.953 [0.945, 0.960]	0.858 [0.846, 0.870]	0.883 [0.870, 0.894]	0.705 [0.689, 0.720]
P2 smoothing only	0.945 [0.935, 0.953]	0.779 [0.762, 0.794]	0.908 [0.897, 0.919]	0.593 [0.575, 0.610]
P2 smoothing only, seed 2	0.953 [0.945, 0.960]	0.833 [0.818, 0.847]	0.921 [0.911, 0.931]	0.629 [0.611, 0.645]
P1	0.952 [0.944, 0.960]	0.807 [0.792, 0.821]	0.885 [0.874, 0.897]	0.569 [0.551, 0.587]
P1 smoothing only	0.965 [0.958, 0.971]	0.871 [0.859, 0.882]	0.925 [0.916, 0.934]	0.673 [0.656, 0.689]
P1 smoothing only, seed 2	0.963 [0.956, 0.969]	0.869 [0.857, 0.880]	0.940 [0.932, 0.948]	0.689 [0.673, 0.705]
P1 smoothing only, seed 3	0.967 [0.960, 0.973]	0.900 [0.889, 0.910]	0.923 [0.913, 0.932]	0.679 [0.663, 0.695]
P2 symmetric smoothing	0.968 [0.961, 0.974]	0.871 [0.858, 0.882]	0.913 [0.902, 0.923]	0.670 [0.653, 0.686]
P2 symmetric smoothing, seed 2	0.950 [0.941, 0.958]	0.796 [0.780, 0.810]	0.935 [0.926, 0.944]	0.646 [0.628, 0.661]
Effort scheme, symmetric smoothing	0.966 [0.960, 0.973]	0.822 [0.808, 0.835]	0.934 [0.925, 0.943]	0.614 [0.596, 0.631]
released Effort fine-tuned, symmetric smoothing	0.941 [0.931, 0.951]	0.805 [0.790, 0.819]	0.920 [0.910, 0.930]	0.705 [0.688, 0.722]
paired difference, OURS (three-way, full data) minus				
Effort [10]	+0.007 [-0.002, +0.018]	-0.090 [-0.109, -0.072]	-0.030 [-0.043, -0.018]	-0.208 [-0.228, -0.187]
Forensics Adapter [11]	+0.004 [-0.006, +0.012]	-0.110 [-0.126, -0.094]	+0.009 [-0.006, +0.021]	-0.163 [-0.184, -0.143]
GenD [12]	-0.015 [-0.023, -0.007]	-0.218 [-0.233, -0.203]	-0.042 [-0.053, -0.032]	-0.326 [-0.343, -0.308]
DFD-HR [13]	-0.031 [-0.038, -0.024]	-0.214 [-0.229, -0.199]	-0.035 [-0.046, -0.024]	-0.196 [-0.213, -0.178]
binary control	-0.017 [-0.023, -0.012]	-0.124 [-0.135, -0.113]	-0.021 [-0.029, -0.013]	-0.145 [-0.157, -0.133]

The second seed of OURS is 0.780 and 0.559 on reenactment. On four Celeb-DF-domain reenactment methods (TPSM, DaNet, MCNet, FaceVid2Vid) OURS scores 0.27–0.37. The three seeds of P0 span 0.09 in the Celeb-DF domain, so differences between arms smaller than that are within the seed spread of the reference there. Four caveats attach to the table. The real videos are the FF++ and Celeb-DF-v2 test reals already used elsewhere in the paper, and Celeb-DF has no identity-disjoint split. Every detector in the table is trained on FF++. Our models were trained on a different crop from the released 256 px DF40 crops; Table S3 gives the size of that handicap on Celeb-DF-v2 and DFD. SBI and LAA-Net are absent because their pipelines need their own crops from full frames.

Reenactment and smoothing across detectors. Over the 13 published detectors scored on both the service and DF40, Spearman correlations of the share of Smoothing-90 renders ranked above their original with reenactment video AUC are $\rho = 0.36$ ($p = 0.23$; FF++ domain) and $\rho = 0.49$ ($p = 0.09$; Celeb-DF domain); with the share accused at the 5%-FPR threshold, $\rho = 0.62$ ($p = 0.024$) and $\rho = 0.31$ ($p = 0.30$). Holm-adjusted over the four, the smallest is $p = 0.096$. Without CADDM the four are 0.24, 0.35, 0.57 and 0.13. Across our seven FF++-only arms (seed 1) the share above own original and FF++-domain reenactment AUC give $\rho = 0.25$ ($p = 0.59$). P1-smooth ranks 2.6% of renders above their original and keeps reenactment AUC at 0.869–0.900. The data do not support a general link between reading smoothed skin as forgery evidence and detecting reenactment.

S10 Fusion of a texture-reading detector with CADDM

Table S17: Score fusion with released weights and no training. Each detector's score is replaced by its quantile among the genuine scores of the same source (for the retouching service: its untouched originals); the fused score is the mean, the minimum or the product of the two quantiles. Video AUC.

	commercial Smoothing-90		untreated video AUC		DF40, FF++ domain		DF40, Celeb-DF domain	
Score	above 5%-FPR thr. (%)	above own orig. (%)	Celeb-DF-v2	DFD	FS	FR	FS	FR
GenD [12] with CADDM [17]								
first member alone	68.7	98.5	0.954	0.977	0.966	0.935	0.947	0.828
CADDM alone	6.3	60.9	0.930	0.918	0.869	0.794	0.788	0.542
fusion, mean	28.7	92.2	0.970	0.964	0.955	0.942	0.914	0.761
fusion, minimum	23.4	78.2	0.962	0.956	0.925	0.922	0.867	0.708
fusion, product	27.6	87.2	0.969	0.966	0.944	0.935	0.895	0.733
DFD-HR [13] with CADDM [17]								
first member alone	64.9	99.0	0.943	0.976	0.982	0.932	0.940	0.698
CADDM alone	6.3	60.9	0.930	0.918	0.869	0.794	0.788	0.542
fusion, mean	25.0	91.5	0.963	0.964	0.964	0.942	0.912	0.673
fusion, minimum	21.8	78.1	0.956	0.955	0.932	0.924	0.866	0.659
fusion, product	24.5	87.2	0.962	0.967	0.951	0.935	0.893	0.658
Effort [10] with CADDM [17]								
first member alone	43.2	95.6	0.930	0.972	0.944	0.808	0.935	0.710
CADDM alone	6.3	60.9	0.930	0.918	0.869	0.794	0.788	0.542
fusion, mean	24.6	86.9	0.965	0.966	0.945	0.881	0.910	0.686
fusion, minimum	22.0	77.4	0.956	0.957	0.915	0.853	0.868	0.658
fusion, product	23.7	84.3	0.963	0.967	0.935	0.869	0.894	0.667
OURS P0 (FF++ only, no beautified views) with CADDM [17]								
first member alone	54.2	98.5	0.938	0.981	0.981	0.906	0.925	0.692
CADDM alone	6.3	60.9	0.930	0.918	0.869	0.794	0.788	0.542
fusion, mean	24.4	91.9	0.961	0.969	0.960	0.939	0.893	0.678
fusion, minimum	20.0	79.1	0.955	0.962	0.929	0.923	0.850	0.665
fusion, product	24.2	87.6	0.961	0.971	0.949	0.933	0.876	0.663

Declared before any fused score was computed: the pairs (GenD, DFD-HR, Effort and P0, each with CADDM), the three rules, and the reading. A fusion counts as useful only if, for the same pair and rule, Smoothing-90 is at most half of the first member's rate, reenactment video AUC is within 0.03 of the first member in both domains, and untreated video AUC is not below the better member by more than 0.01; one rule meeting this for at least two of the three published pairs counts. No rule does (Table S17). Under the mean rule GenD fails the reenactment condition in the Celeb-DF domain (-0.067); DFD-HR and P0 fail the untreated condition on DFD (-0.012 against 0.010); Effort fails the halving condition by 3 points. The minimum rule cuts the accusation further (20.0–23.4%; product 23.7–27.6%), and both lose more reenactment in the Celeb-DF domain (0.66–0.73). The ensemble of the alpha-blending study [19], which combines a blending detector with blend-resistant models, is the closest prior art; our reading differs in that it asks whether fusion changes the ranking of a smoothed face against its own original, and it does not (87–92% of pairs under the mean rule).

S11 Methods that did not meet their bars

Table S18: Every method arm tried under a pre-registered bar. FF++ only unless noted; each block starts with its reference. Bars for the FF++-only arms: Celeb-DF-B frame $\text{AUC}_b \geq 0.890$ and $\Delta\text{MISS}_{\text{beauty}} \leq 7.5$, untreated Celeb-DF-v2 AUC within 0.01 of the reference. None is met. "Random edits" is a random-augmentation control, not an adversarial arm.

	Celeb-DF-B				Celeb-DF-v2	DFD filters
Arm	AUC	AUC_b	$\Delta\text{MISS}_{\text{beauty}}$	FA_b	AUC untreated	AUC_f
random colour grading (reference)	0.938	0.872	+7.5	7.8	0.898	0.932
no grading	0.929	0.849	+14.1	7.2	0.880	0.917
operator, random edits	0.932	0.876	+7.9	7.7	0.873	0.932
operator, worst of 4 draws	0.928	0.867	+12.9	5.6	0.860	0.935
operator, 1-step search	0.930	0.864	+8.2	8.3	0.876	0.935
operator, search within measured edits	0.928	0.869	+15.1	5.2	0.874	0.934
operator, consistency objective	0.924	0.856	+13.0	5.7	0.849	0.922
operator, 3-step search	0.922	0.859	+8.9	7.6	0.849	0.937
decision pooled from the evidence map	0.937	0.877	+14.9	5.0	0.889	0.924
same, no beautified views	0.942	0.891	+14.7	4.4	0.899	0.932
full data, binary (reference)	0.949	0.896	+7.0	7.4	0.914	0.952
full data, binary, operator search	0.942	0.892	+9.7	6.0	0.911	0.945
released Effort, mapped (reference)	0.909	0.808	+11.2	4.7	0.886	0.904
released Effort + 2 epochs, no beautified views	0.903	0.805	+5.8	9.4	0.866	0.918
released Effort + 2 epochs, both classes	0.895	0.803	+2.7	10.3	0.861	0.916

The operator. A differentiable edit $B(x; u)$ on a face crop in $[0, 1]$ with every component bounded by a sigmoid: per-channel gamma, gain and lift; a 3×3 colour matrix close to the identity; saturation from monochrome to 1.7; contrast about the mean; a 4×4 luminance field and a per-channel tint field, upsampled bilinearly; a blend toward a Gaussian-smoothed copy; additive grain. Its parameters were fitted by gradient descent to 1,000 pairs each of four measured families (Table S19); the search of the "measured edits" arm starts from a fitted edit and is projected onto the 0.5th–99.5th percentile range of the fitted parameters.

Table S19: The operator fitted to measured edits (training split only). The residual of the commercial renders is geometry, which a photometric operator cannot express.

Measured edit	n	PSNR before	PSNR after fit	10th pct
Tencent retouching	1000	21.4	28.6	24.8
Megvii retouching	1000	29.1	30.6	28.0
training colour filters	1000	21.0	40.1	32.0
random grading	1000	24.2	48.4	39.0

The search works; the training does not carry over. On 400 FF++ validation frames the reference model keeps 96.0% of genuine faces genuine without an edit, 71.0% under a random draw of the operator, 41.0% under the worst of four draws, 8.5% under a one-step signed-gradient search and 0.0% under ten steps; forged faces stay forged in 93.5, 89.5, 83.0, 74.0 and 20.0% of cases. After training on the one-step worst case the genuine accuracy under that search is 71.5% (25.0% under ten steps), and after the three-step variant 76.5% (41.5%). The gain does not carry over to the app presets or to filters from a library the operator was not fitted to, and the forged accuracy under random edits falls from 89.5 to 79.5%: with a label loss, edited views of genuine faces dominate the signal and the model learns that a heavily edited face is less likely forged.

A learned inverse. The exact algebraic inverse of the invertible part of the operator restores an edited frame to 53–58 dB when the true parameters are given. A ResNet-18 estimator trained to predict them from the edited frame alone reaches 26.7 dB on random edits (25.9 before) and 22.7 dB on the held-out filters (21.0 before), and moves untouched frames to 34.4 dB. In front of the detector it changes the false alarms under the held-out filters from 21.5 to 23.6%. A single image does not determine its colour transform.

Edit sensitivity as evidence. The response of the forged logit to six fixed probe edits is larger for genuine than for forged frames on average, but inside the deciles of the clean logit where both classes occur the two respond alike (1.15 vs 0.96; 0.48 vs 0.10), so the asymmetry is a ceiling effect; adding the responses to the logit leaves the false alarms under filters at 16–22%.

A decision pooled from the evidence map. With a zero map target on every beautified item and the forged logit taken as the mean of the top 5% map cells, the map stays dark on filtered genuine faces (mean logit -10.2 against $+1.9$ for the edit map) and the false alarms under filters fall from 21.5 to 14.7%, but the beautification-specific miss rise on Celeb-DF-B grows from 7.5 to 14.9 points.

Fine-tuning a released checkpoint. The released Effort weights were loaded into our pipeline (the mapped model reproduces the released scorer: Celeb-DF-B frame AUC 0.909, AUC_b 0.808) and fine-tuned for two epochs. With both-class exposure the miss rise falls from 11.2 to 2.7 points, the false alarms on beautified genuine faces rise from 4.7 to 10.3%, AUC_b stays at 0.803 and the untreated AUC falls by 0.014; without beautified views the same fine-tuning gives 5.8 points and 9.4%. The operating point moves; separability does not.

S12 The beautified class of the three-way models

The paper reads every model by $P(\text{fake})$ alone. Table S20 gives the native three-way reading of the models that have a beautified class; its numbers are not comparable with the one-rule numbers.

Table S20: Retouching service, native three-way decision: 2,210 FFHQ originals and their renders (four operations at three levels, 26,519 images). Paired rank: share of pairs in which the render gets a higher $P(\text{filter})$ than its own original (50 means no signal).

	originals	retouched (pooled)				paired rank at level 90	smoothing-90
Model	real/fake/filter	→filter	→fake	balanced	paired rank	eye / white / lift / smooth	→fake
FF++ c23 training only (zero-shot)							
three-way, observed forgeries	36.8/5.7/57.5	66.5	9.1	54.5	79.8	97 / 89 / 79 / 59	27.3
three-way, part edits + evidence head	16.7/55.9/27.4	28.6	59.4	50.6	60.9	89 / 78 / 60 / 20	85.8
Trained with two vendors' renders and their single-operation components; Alibaba never seen (ours)							
OURS (CLIP ViT-L/14 + LoRA)	78.4/0.8/20.8	67.7	0.6	73.5	92.9	98 / 92 / 94 / 99	0.8
OURS-lite (EfficientNet-B4)	81.0/0.4/18.6	63.8	0.3	72.6	93.8	99 / 91 / 93 / 99	0.4

Three-way models trained on FF++ alone separate retouched from untouched faces at chance and call commercial smoothing fake, up to 86% of faces at the strongest level. Trained with the single-operation components of two other services, OURS calls 0.6% of the retouched faces fake by its own three-way decision (under the one rule 4.7%), sends 67.7% of them to FILTER and gives 92.9% of renders a higher filter score than their own original. It also sends 20.8% of the untouched originals to FILTER, so its balanced accuracy on "retouched or not" is 73.5% (75.4% for the second seed), against a pre-registered target of 75%. The AUROC of $P(\text{filter})$ between originals and renders is 0.80–0.82 for every model trained with components. The symmetric-view variant sends 13.9% of the untouched originals to FILTER and has a balanced accuracy of 73.4%. On Celeb-DF-B, OURS sends 13.2% of untreated genuine frames, 47.4% of beautified genuine frames, 16.6% of beautified deepfakes and 3.6% of untreated deepfakes to FILTER (second seed 20.7, 56.9, 16.1 and 3.9%; variant 14.2, 48.1, 17.4 and 3.7%). The attribute recognises about half of the app-beautified genuine frames and labels about one in seven untouched ones as beautified, so it is reported with these error rates and not as a separation result.

Under the one rule the accusation concentrates on smoothing. Table S21 gives the share of renders above the 5%-FPR threshold of the originals per operation.

Table S21: The retouching service per operation under the one rule: share of renders (%) above the threshold that accuses 5% of the untouched originals. Level 90 for the four operations, and the three levels of smoothing.

	level 90			smoothing		
Detector	whitening	face slimming	eye enlarging	level 30	level 60	level 90
Xception [2]	5.7	3.7	4.6	16.6	36.2	51.6
EfficientNet-B4 [3]	5.1	4.0	4.7	11.5	23.9	36.3
F3Net [4]	5.4	4.2	4.4	15.0	33.2	50.6
SPSL [5]	5.2	4.9	4.9	14.9	34.1	51.2
UCF [6]	6.6	4.0	4.2	16.8	32.5	41.4
RECCE [7]	5.3	4.1	4.5	11.9	22.7	36.8
CORE [8]	5.7	4.4	4.8	14.8	36.6	54.8
SRM [9]	4.5	3.8	4.7	18.7	45.0	62.8
SBI [14]	5.4	6.2	5.5	14.8	31.6	37.3
UnivFD‡ [15]	6.2	5.7	11.7	6.8	12.3	16.6
NPR‡ [16]	n/a	n/a	n/a	n/a	n/a	n/a
Effort [10]	11.1	9.6	16.8	13.2	27.2	43.2
Forensics Adapter [11]	5.4	6.2	7.0	10.2	25.8	39.7
GenD [12]	8.1	6.4	7.5	20.7	55.3	68.7
DFD-HR [13]	7.7	8.0	9.6	17.2	46.7	64.9
CADDM [17]	4.7	6.0	5.9	5.4	5.6	6.3
LAA-Net [18]	5.2	4.8	4.8	11.5	17.2	17.2
OURS, three-way, full data	4.8	3.7	3.6	6.2	5.0	8.6
OURS, binary control	5.4	5.0	4.3	5.6	6.3	9.1
OURS, symmetric-view variant	6.3	3.0	3.3	7.3	5.3	8.1
OURS, FF++ only, P0	6.8	5.8	5.4	17.0	40.8	54.2
OURS, FF++ only, P2 symmetric smoothing	5.8	5.0	5.0	4.5	3.9	5.9

Table S22: All trained variants on the service (2,210 originals, 26,519 renders). Columns as in Table S20.

	originals	retouched (pooled)				paired rank at level 90	smoothing-90
Model	real/fake/filter	→filter	→fake	balanced	paired rank	eye / white / lift / smooth	→fake
FF++ c23 training only (zero-shot)							
three-way, RepViT-M0.9	0.8/2.5/96.7	92.6	6.4	47.9	48.4	73 / 55 / 52 / 15	17.7
three-way, observed forgeries	36.8/5.7/57.5	66.5	9.1	54.5	79.8	97 / 89 / 79 / 59	27.3
three-way, observed and self-blended forgeries	33.8/16.7/49.5	54.6	22.1	52.6	69.2	92 / 83 / 65 / 44	50.2
same, faithful blend	28.2/22.1/49.6	52.8	28.0	51.6	65.7	92 / 77 / 62 / 35	58.4
same, label-symmetric degradation	31.7/26.3/42.0	45.3	33.6	51.7	64.3	89 / 76 / 63 / 35	63.1
three-way, part edits + evidence head	16.7/55.9/27.4	28.6	59.4	50.6	60.9	89 / 78 / 60 / 20	85.8
Variants trained with two services' renders (each read once on the test service)							
renders only (EffNet-B4, s1)	93.4/0.8/5.8	38.1	0.6	66.1	92.8	99 / 85 / 95 / 100	0.1
renders only (RepViT)	92.8/1.4/5.9	36.1	1.1	65.1	92.6	96 / 89 / 93 / 100	0.3
renders + single-operation components (EffNet-B4, s1)	76.8/0.5/22.8	68.2	0.3	72.7	94.2	99 / 90 / 94 / 100	0.4
renders + single-operation components (EffNet-B4, s2)	66.6/0.3/33.2	74.9	0.2	70.8	94.2	99 / 92 / 93 / 99	0.4
renders + single-operation components (RepViT)	43.4/0.9/55.7	83.7	0.5	64.0	95.1	99 / 91 / 96 / 99	1.1
components below noise floor removed (EffNet-B4, s1)	76.7/0.4/22.9	68.8	0.3	72.9	94.9	99 / 92 / 94 / 99	0.5
same + evidence head (EffNet-B4, s1)	83.2/0.6/16.2	62.2	0.5	73.0	94.1	99 / 91 / 93 / 99	0.8
same + part-level forgeries (lite) (EffNet-B4, s1)	81.0/0.4/18.6	63.8	0.3	72.6	93.8	99 / 91 / 93 / 99	0.4
same + evidence head, CLIP (CLIP-L/14, s1)	77.4/0.4/22.2	71.4	0.2	74.6	93.8	98 / 90 / 94 / 100	0.2
same + evidence head, CLIP (CLIP-L/14, s2)	73.0/0.3/26.7	73.6	0.2	73.4	95.1	99 / 94 / 96 / 100	0.2
same + part-level forgeries, CLIP, filter-F1 selection (superseded) (CLIP-L/14, s1)	57.9/0.3/41.9	82.3	0.1	70.2	94.8	98 / 94 / 95 / 100	0.0
same + part-level forgeries, CLIP (CLIP-L/14, s1)	89.5/0.6/9.8	59.2	0.5	74.7	95.5	98 / 94 / 96 / 100	0.1
same + part-level forgeries, CLIP (CLIP-L/14, s2)	83.3/0.4/16.2	67.5	0.2	75.6	93.0	98 / 88 / 95 / 100	0.2
main recipe without colour grading (CLIP-L/14, s1)	82.1/0.6/17.3	65.7	0.3	74.2	92.8	97 / 89 / 93 / 99	0.4
main recipe without service edits on forgeries (CLIP-L/14, s1)	83.8/0.4/15.8	65.6	0.2	74.9	94.7	98 / 94 / 94 / 100	0.1
main data, three-way softmax head (CLIP-L/14, s1)	81.5/0.7/17.8	66.4	0.5	74.3	94.1	98 / 94 / 93 / 99	0.5
main model (OURS) (CLIP-L/14, s1)	78.4/0.8/20.8	67.7	0.6	73.5	92.9	98 / 92 / 94 / 99	0.8
main model (OURS) (CLIP-L/14, s2)	79.8/0.5/19.6	70.5	0.4	75.4	94.3	98 / 92 / 95 / 99	0.6

S13 Explanations: all variants

Table S23: Restoration test on held-out part-level forgeries for every variant. Columns as in Table VII.

	donor transplant				SD-1.5 inpaint				SDXL inpaint (unseen editor)			
Map	IoU_7	flip	ceil.	compl.	IoU_7	flip	ceil.	compl.	IoU_7	flip	ceil.	compl.
centre prior / random region (last col.)	0.310	38.6	–	17.0	0.249	40.9	–	17.9	0.255	40.2	–	17.7
no head, Grad-CAM [20]	0.367	50.9	85.8	15.2	0.259	45.4	85.2	25.0	0.269	48.0	85.9	23.6
no head, Grad-CAM++ [21]	0.414	51.7	85.8	7.2	0.327	47.8	85.2	16.7	0.330	48.8	85.9	19.1
no head, Score-CAM [22]	0.408	56.5	85.8	5.9	0.334	52.4	85.2	15.4	0.341	52.4	85.9	17.1
head, part edits, FF++ only	0.723	84.0	84.7	0.4	0.752	82.6	83.2	0.7	0.746	82.9	83.8	0.7
+ symmetric degradation	0.723	86.3	87.1	0.4	0.719	82.5	83.7	2.0	0.728	83.7	84.4	1.5
+ restoration-consistency loss	0.759	89.4	90.0	0.4	0.614	75.2	79.8	4.4	0.684	82.5	84.6	2.6
+ consistency + symmetric degradation	0.722	88.4	89.3	0.7	0.723	85.2	86.1	2.4	0.732	86.1	86.9	1.8
head, vendor data, no part edits (EffNet-B4)	0.418	61.4	82.4	12.0	0.280	39.5	63.5	23.4	0.315	46.2	72.3	22.5
head, vendor data, no part edits (CLIP)	0.479	78.8	89.4	2.5	0.376	53.9	69.1	14.4	0.445	68.9	78.3	8.4
Ours-lite (EffNet-B4)	0.807	87.9	88.0	0.0	0.820	87.4	87.6	0.4	0.812	86.9	87.2	0.4
full-data CLIP model, filter-F1 selection (superseded)	0.832	91.7	91.9	0.1	0.845	91.5	91.6	0.2	0.827	90.9	91.2	0.5
Ours (CLIP ViT-L/14 + LoRA)	0.834	94.0	94.1	0.1	0.841	93.6	93.7	0.4	0.831	93.5	93.6	0.3

The donor-transplant test set uses donors from test videos only. A first build of that set drew donors from training videos (1,023 donor-frame and part pairs occurred in training); it was rebuilt, and the numbers of Ours moved by less than 0.2 points. Figure S2 shows examples. The test is a localisation test plus a causal check that restoring the localised region removes the detection; it does not compare maps across models, evaluate missed forgeries or false accusations of genuine faces, or use a non-oracle infill, and these controls are left for future work.

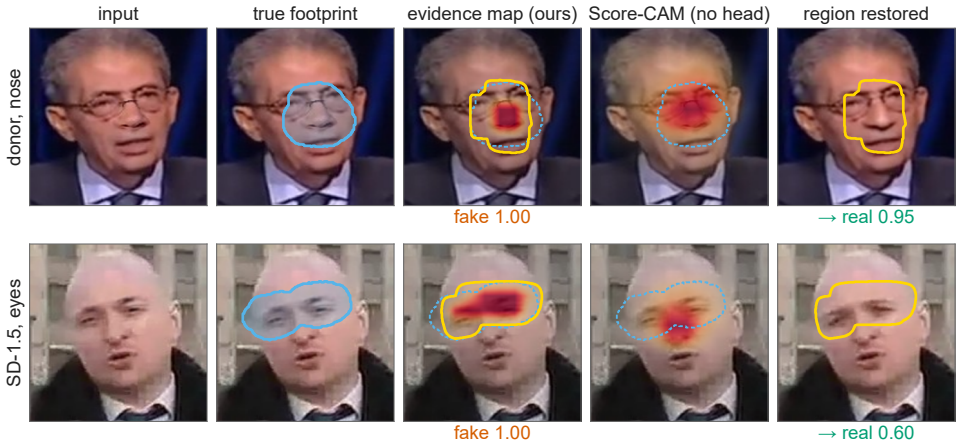


Figure S2: Held-out part-level forgeries: input, exact footprint (blue; dashed on the maps), evidence map with the three-way decision, Score-CAM on the same backbone without the head, and the result of restoring the region cut from the evidence map (yellow, same area as the footprint) to the original, with the decision on it.

S14 Sources that were examined and excluded

AR beauty filters. On paired AR-filter images, a model trained with such pairs called 94.8% of the filtered images *filter* with the face hull greyed out: the capture pipeline, not the face, carries the label. No claim is made on AR filters. **WildDeepfake.** The 800 test frames available to us come from six sequences, which gives intervals of 0.3–0.5 AUC; not reported. **DFDC, DFDCP.** Not available to us in a form with genuine and forged frames from the same pipeline.

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